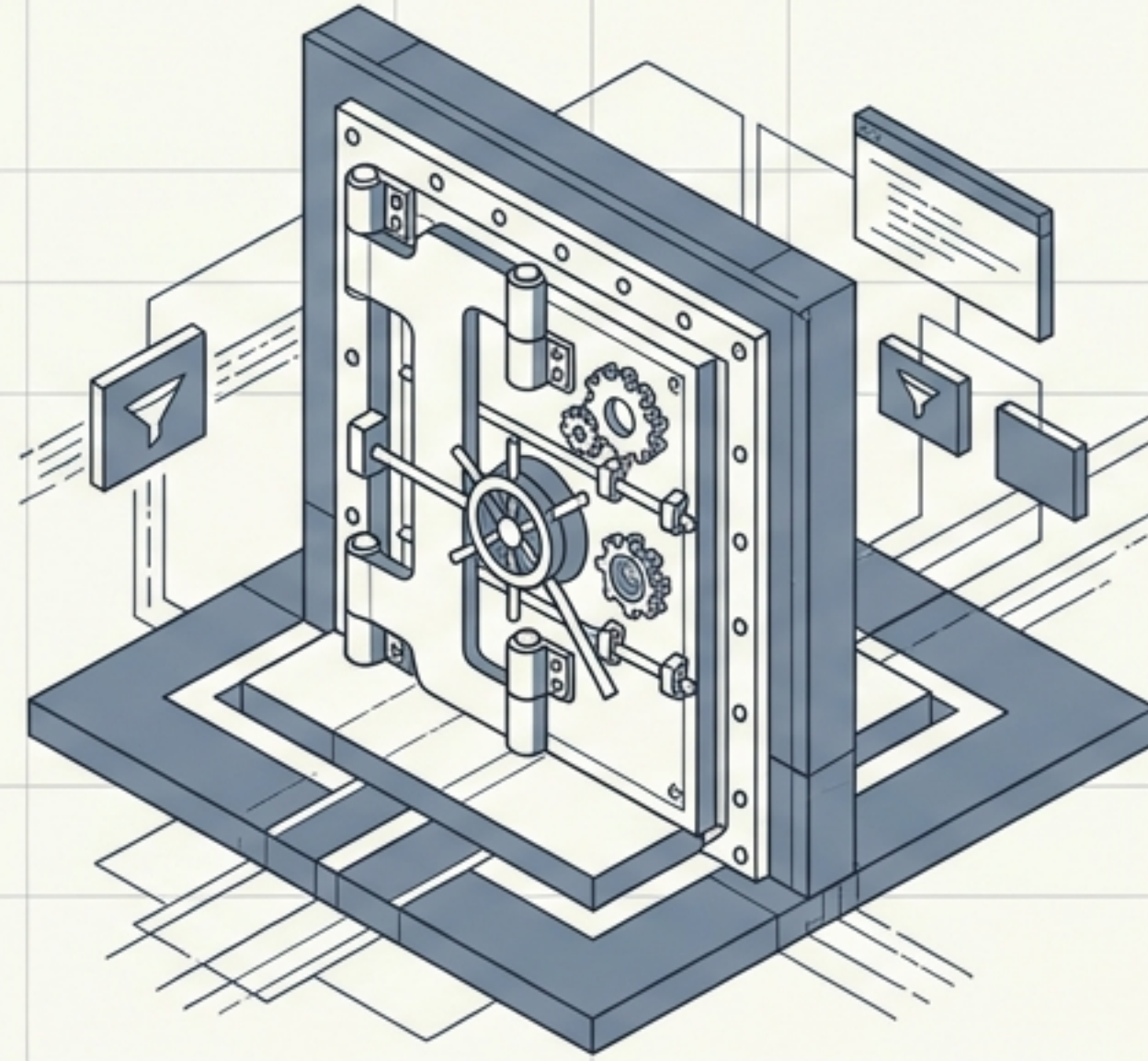


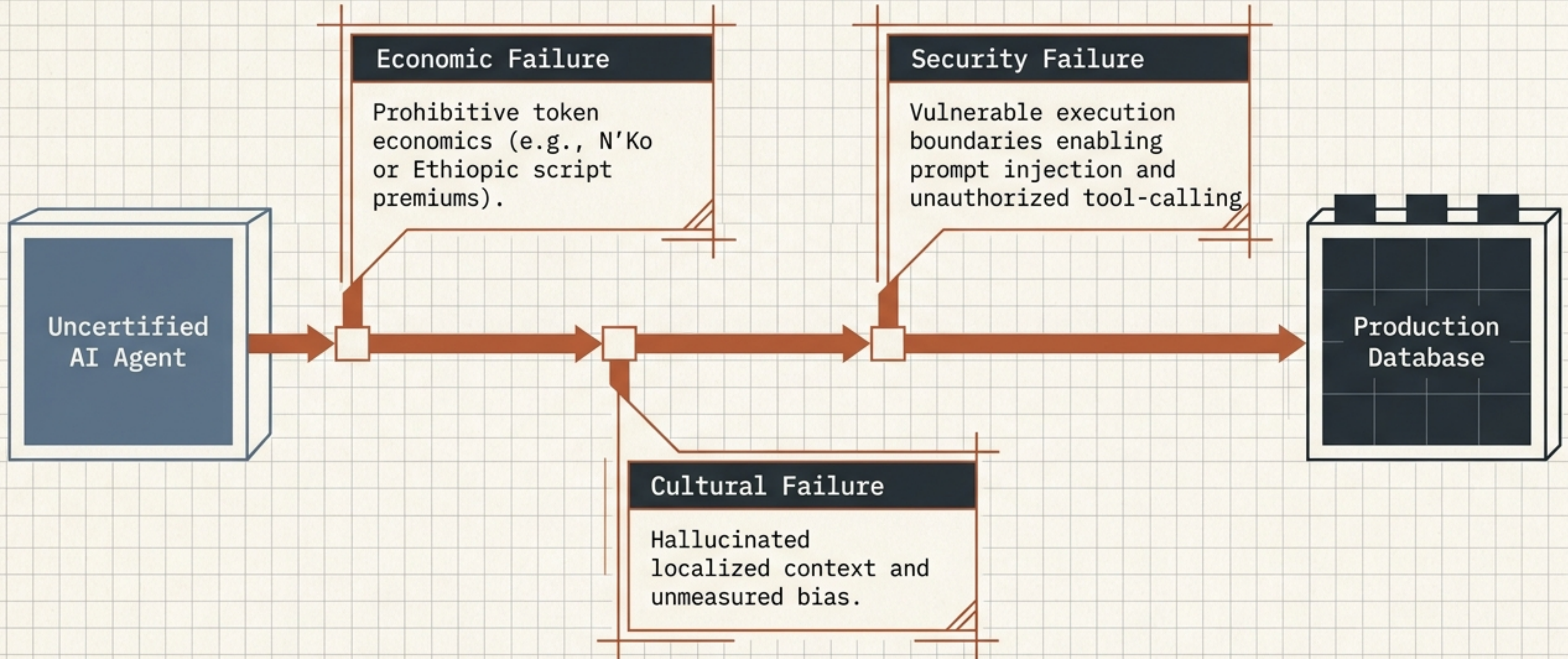


The the AfrEval Tollbooth

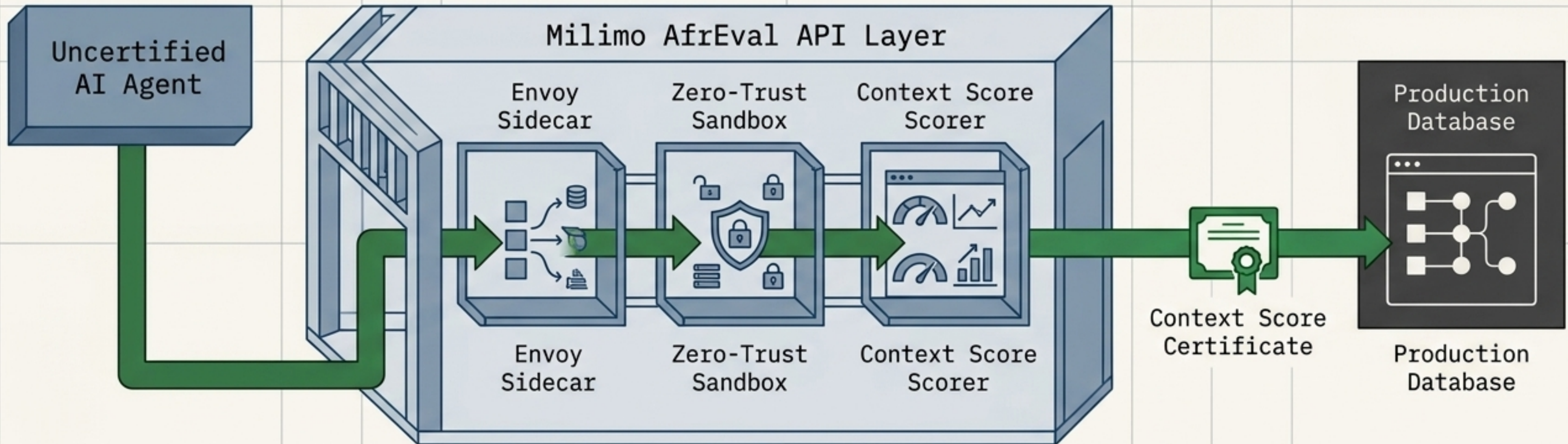
Autonomous, context-aware AI alignment
and benchmarking for Africa.

Ensuring AI is safe, viable,
and localized before it
touches production.

The Baseline Vulnerability



The Enterprise Tollbooth



The Enterprise Tollbooth

An API layer that routes every agent through a certification pipeline before it touches production.

The Guarantee

Issues a Context Score proving the model is secure, aligned, and economically viable for African deployment.

The Context Score Quadrant

The four primary guarantees of the AfrEval certification.

1. Linguistically Accurate

Script-aware tokenization and robust edge ASR (Automated Speech Recognition) baselines.

2. Culturally Safe

BiasScope probing and autonomous bias-correction alignment via live judge backends.

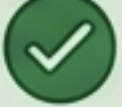
3. Economically Viable

Structurally lowering token premiums for African languages to match standard Latin-script baselines.

4. Functionally Secure

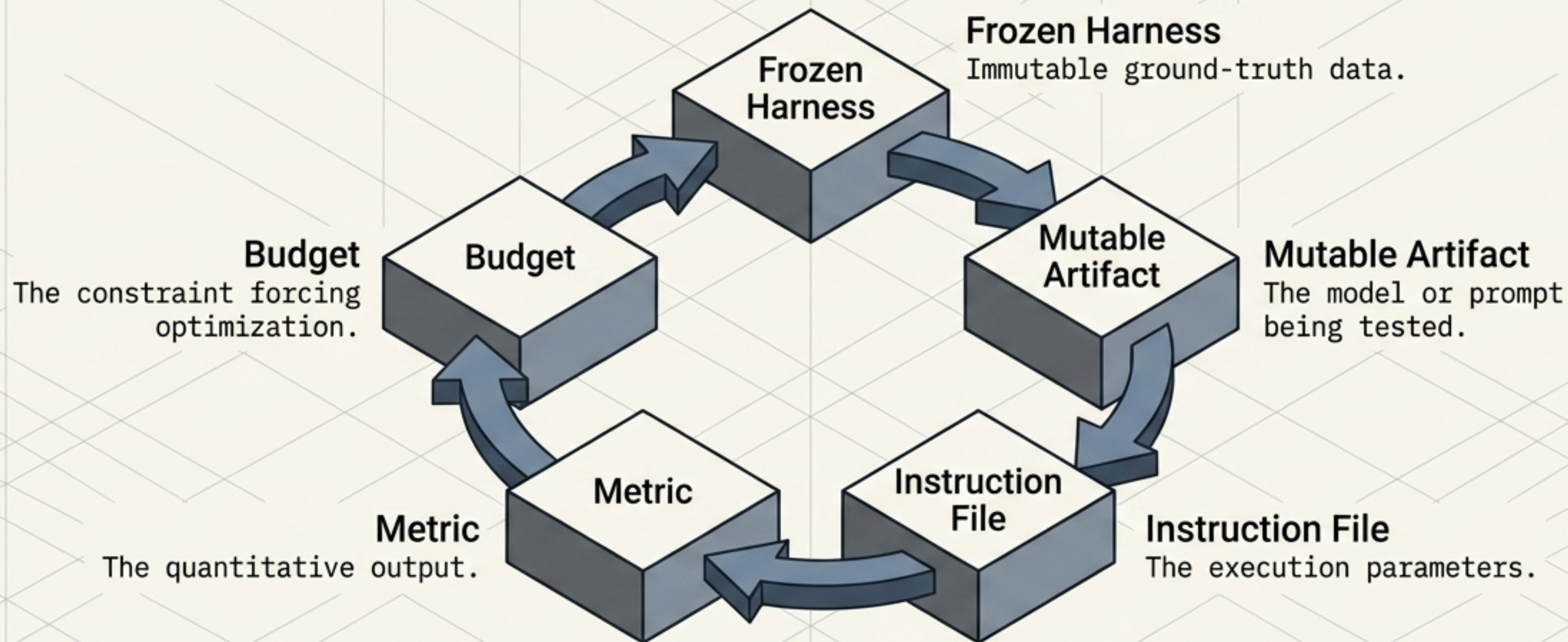
Execution bounded by four defensive seams, JWS HS256 validation, and strict replay protection.

Architectural Comparison

	Standard AI Deployment	AfrEval Certified
Economics	 Generic tokenizers with heavy African language premiums.	 Optimized BPE candidates minimizing script premiums.
Bias & Alignment	 Untested cultural alignment and hallucinated context.	 Autonomous judge-bias probing via open-source Ollama.
Security	 Open tool-calling parameters and vulnerable boundaries.	 4-seam JWS HS256 validation + ReplayGuard protection.
Compliance	 Outdated, Western-centric legal frameworks.	 7/7 citation-current African legal frameworks (AU, ECOWAS, etc.).

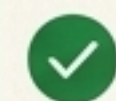
The Theoretical Engine

The Generalized Karpathy Loop driving autonomous system alignment.



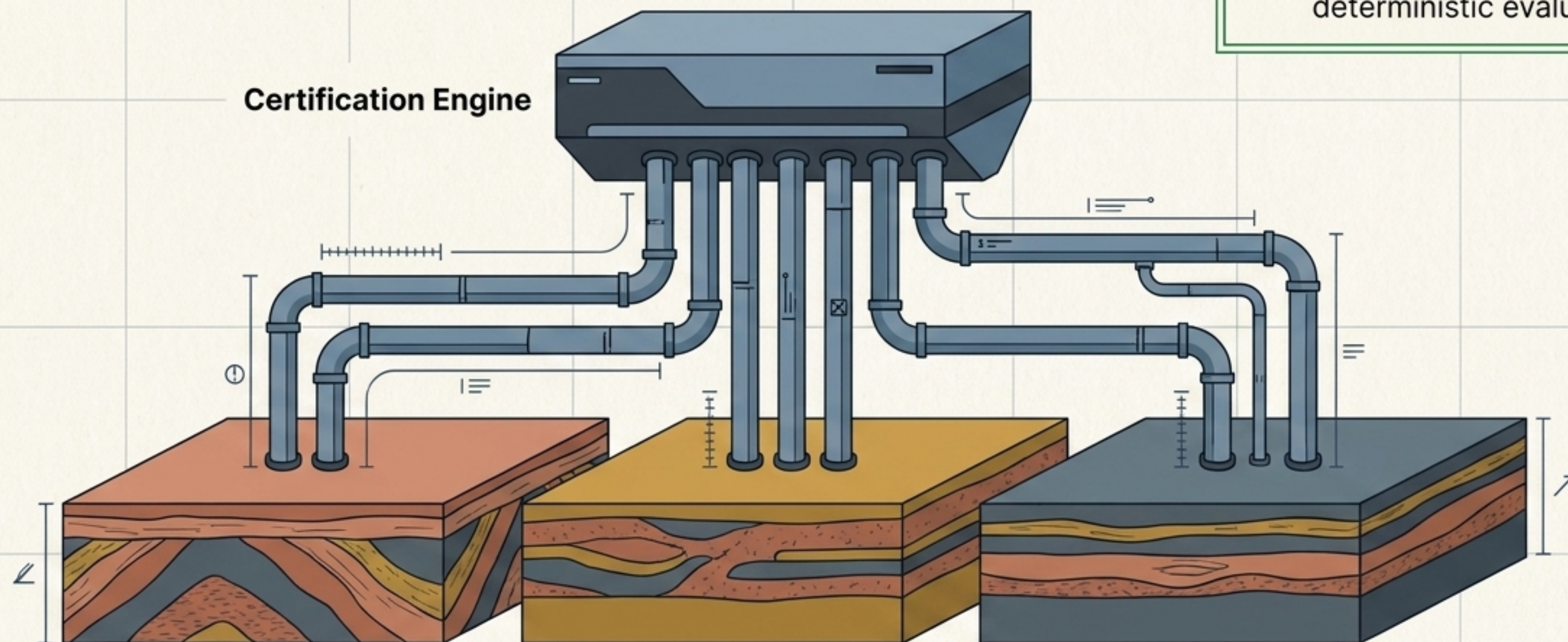
Every subsystem (tokenizer, bias, compliance) operates as an autonomous loop over frozen data substrates, continuously driving toward optimization without human bottlenecking.

The Foundation: Frozen Data Substrates



Phase 0 Closed:

All three harnesses frozen and checksummed for deterministic evaluation.



WAXAL

The ultimate audio substrate.
72,145 QA-approved audio clips
across 18 languages.

AfroBench

The comprehensive language
evaluation harness vendored
from McGill-NLP.

afri-fertility

Specialized domain
benchmarking for high-stakes
healthcare contexts.

Telemetry: Accuracy & Economics

Structural Economics



SIB-200 score jumps from 37.25 to 62.82.

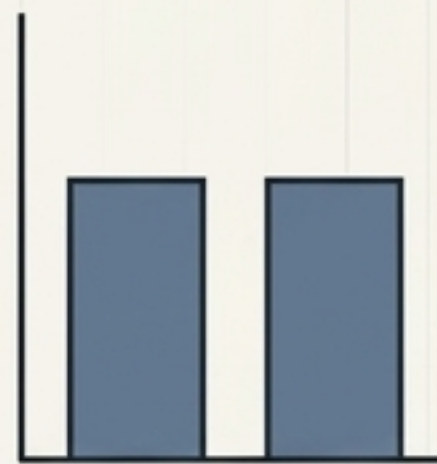
Token Premium Reduction



Ethiopic script premium slashed from 7.83 to 3.38.

Achieved with zero English-CPT regression. N'Ko third number accurately measured at 1.53.

Edge ASR Baseline



Zero-shot baseline ASR established at 41.0% macro-WER.

Latin-African linguistic gap successfully narrowed.

Telemetry: Security & Culture



Airlock Hardening

31 attack variants tested. 0 bypasses.
Seam-4 replay protection active (grant jti nonce + optional exact-call binding).



BiasScope Alignment

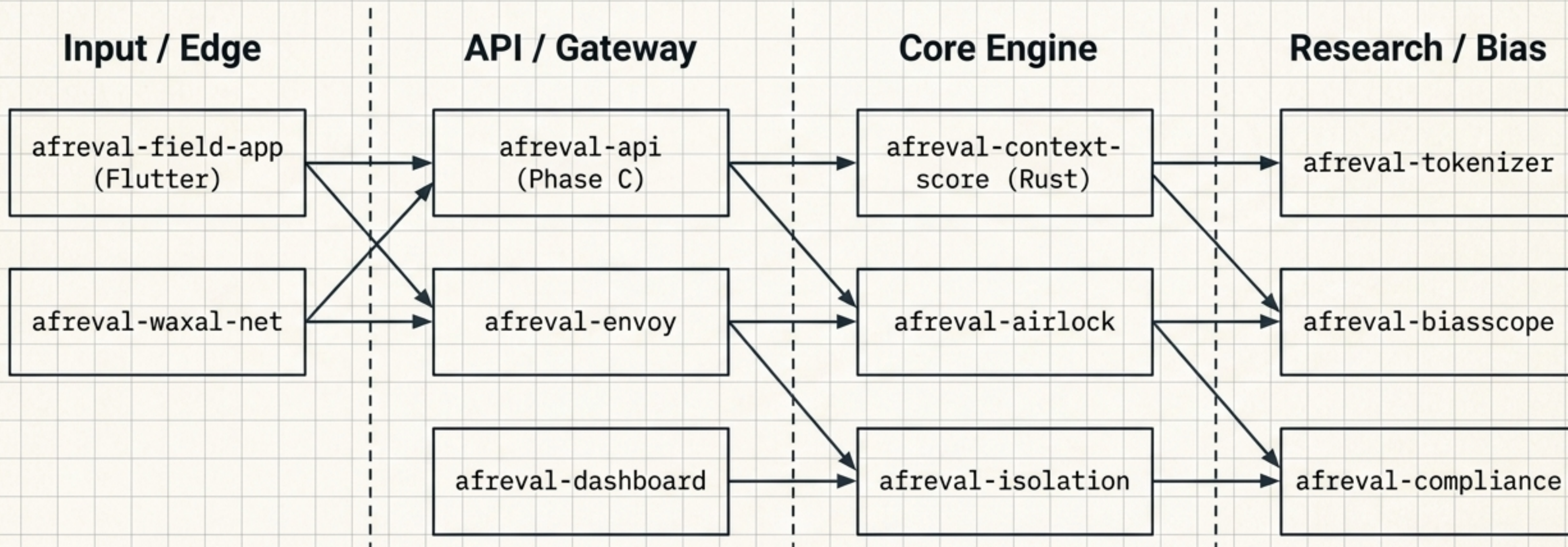
Live run via open-source Ollama judge proves a genuine bias delta (1.0 gap between English rejected / African accepted).
Bias-correction is actively wired directly into certification.



Legal & Framework Compliance

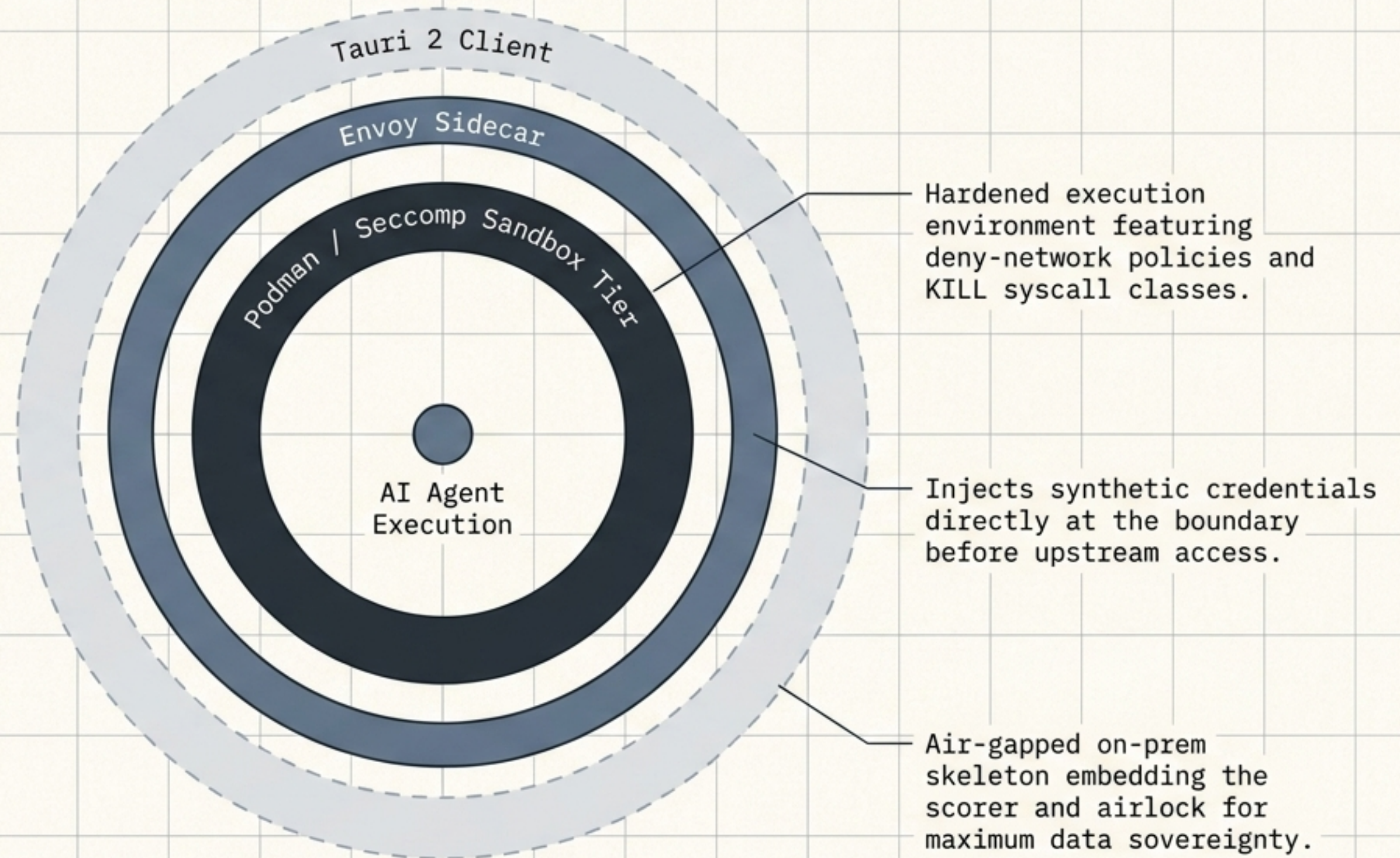
100% (7/7) citation-current with AU/Malabo, ECOWAS/SADC, Kenya ODPC, and Nigeria NDPC.

The Architectural Pipeline

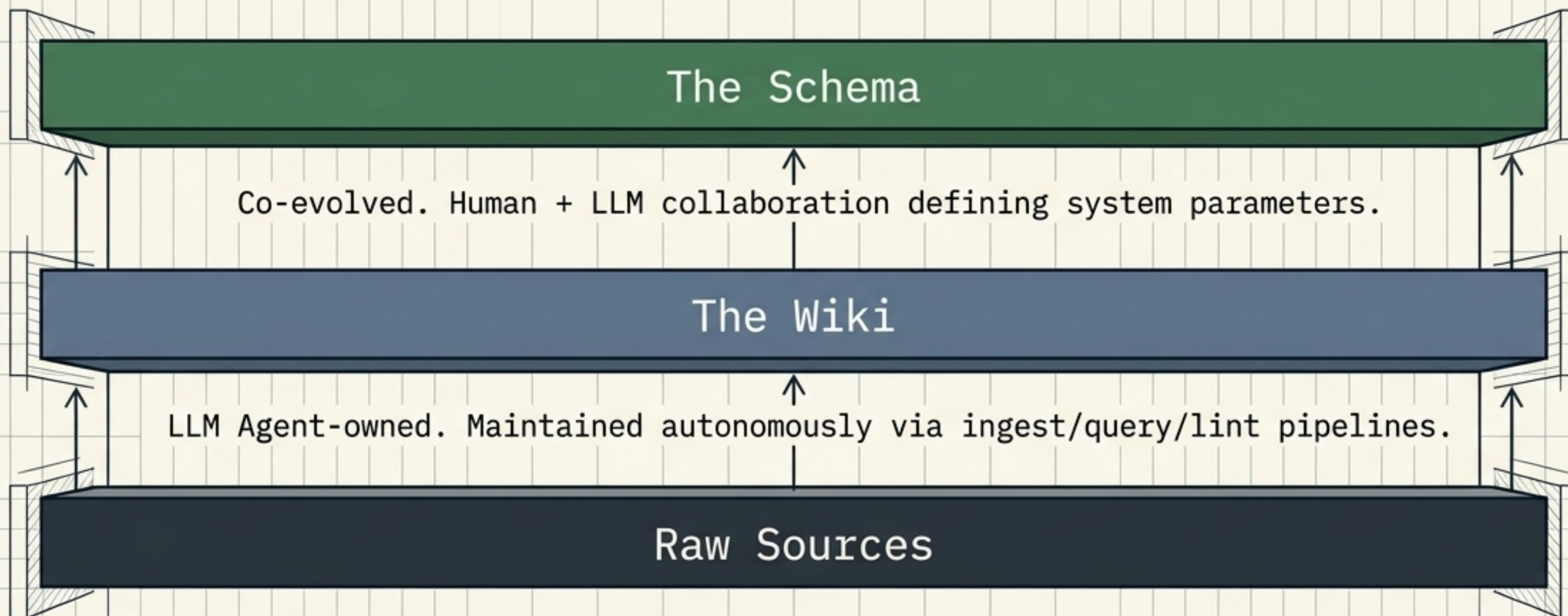


11 unified repositories powering a single, seamless certification pipeline.

Enterprise Isolation Architecture



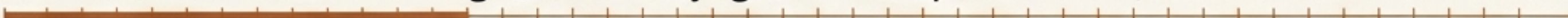
The Governance Model



Human-owned. Immutable dev-docs and vendored substrate repositories.

A resilient, self-maintaining governance model blending human immutability with autonomous LLM upkeep.

System Readiness Status

	COMPLETED OPERATIONAL COMPONENTS
	Phase 0: Data Harnesses & Substrates Live - COMPLETE 
	Phase 1: Core Scorer & Tokenizer Complete - COMPLETE 
	Phase 2: Airlock, Bias Probing, and Compliance Complete - COMPLETE 
	Phase C: API Layer Fully Operational - COMPLETE 
	IN-PROGRESS OPERATIONAL DEPLOYMENT
	Phase 4: ASR fine-tuning (currently gated on upstream HF) - IN PROGRESS 
	Phase 5: On-prem client skeleton prep - PREP 

The foundational architecture and core engine are fully live and deterministic. Remaining work is strictly operational deployment.

Developer Integration

SDK Integration

```
from afreval.client import AfrEvalClient

# Initialize the client
client = AfrEvalClient(api_key="...")

# Call the Phase C certification API
certification_result = client.certify_agent(
    agent_id="agent_v3.2",
    config={"phase": "C", "strict": True}
)

print(certification_result.status)
# Output: CERTIFIED
```

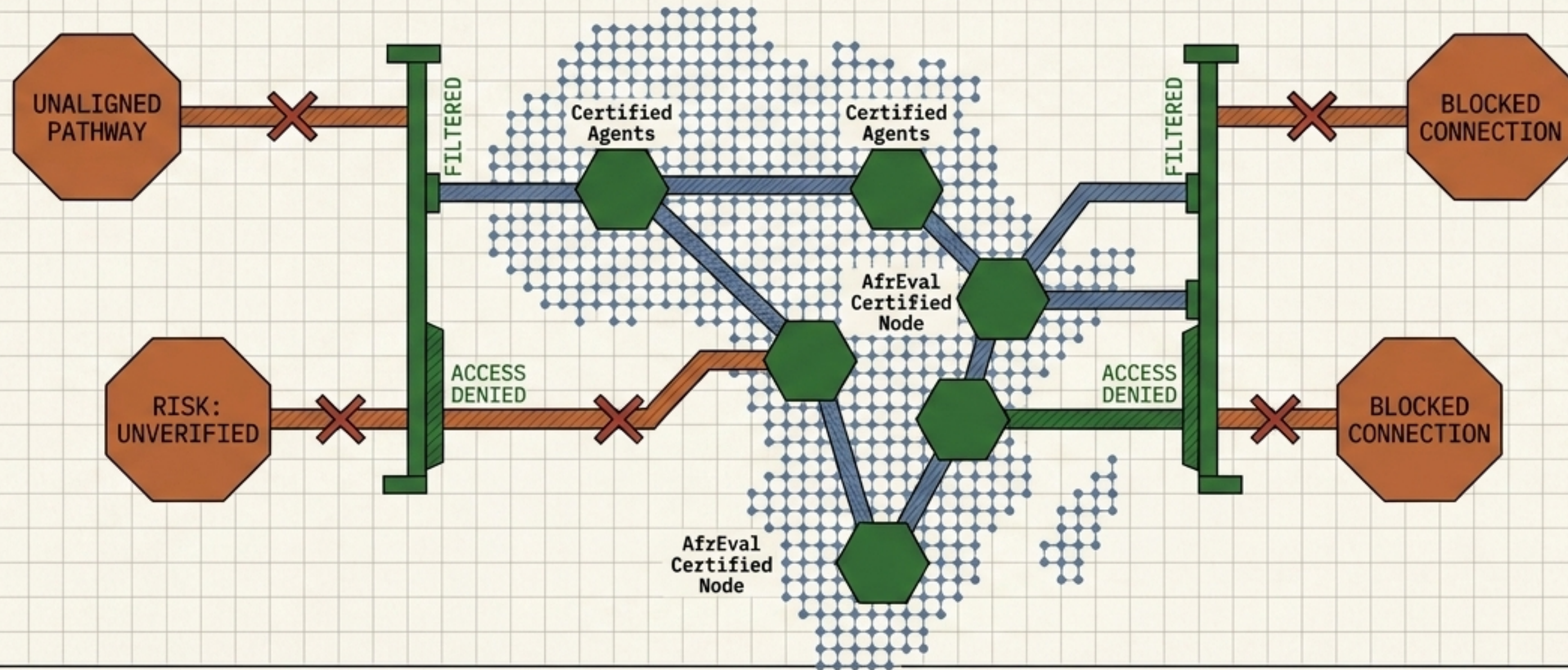
Packaged Python and TypeScript client SDKs are live and ready to call the Phase C certification API.

Deterministic Core

sha256: 4b2a1c... **validated** ✓

The underlying Rust scorer guarantees bit-identical repeated runs, issuing auditable, cryptographically secure certs for every validated agent.

The Future of African AI



AfrEval is not a static benchmark; it is living enterprise infrastructure.

By routing agents through an autonomous, context-aware tollbooth, we guarantee the next generation of AI in Africa is secure, economically viable, and deeply aligned with local realities.